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Perceived Working Conditions and Intention to Adopt Digital Safety Training in High-Risk Productive Sectors: An Exploratory Study in Manufacturing and Agriculture in Northwest Italy

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Abstract: Agriculture and manufacturing report the highest rate of occupational accidents and fatalities in Italy. Safety training provided through digital devices has been shown to be effective in promoting safety behaviors at work. This study aimed to investigate through a questionnaire the perceptions of working conditions, risks in using machines, and interest in using digital devices for safety training purposes in a group of vineyard workers (VWs, N = 40) and manufacturing workers (MWs, N = 39) in Northwest Italy. Referring to working conditions, VWs significantly differ compared to MWs ($p < 0.05$) in fatigue perception, repetitiveness, quantity and definition of tasks compared to the available time, work pace definition, and level of communication. Tractors and lathes were considered the most hazardous machinery for VWs and MWs, respectively. For both groups, workers' age negatively correlated with digital device use ($r = -0.399$ $p < 0.05$ for VWs, $r = -0.673$ $p < 0.01$ for MWs) but not with interest in using them. Device adoption positively correlated with the perceived importance of gamification content ($r = 0.193$ and $r = 0.164$, $p > 0.05$ for VWs and MWs, respectively), but the video lessons reported a higher mean score by both groups as preferred content. These findings suggest that digital safety training requires customized content to effectively adapt to different productive sectors.

Keywords: agriculture; manufacturing; risk perception; tractor; industrial machinery; digital devices; digital contents; safety training

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1. Introduction

According to the more recent statistics, more than 395 million workers globally suffered from non-fatal work injury in 2019 and nearly 3 million workers lost their lives due to work-related causes [1]. The most hazardous sectors include manufacturing, forestry, mining, construction, agriculture, and fishing, which account for approximately 200,000 fatal injuries per year, representing 63% of all occupational fatalities [1]. Specifically, in Europe, manufacturing accounted for 15.2% of all fatal accidents at work in 2022, while agriculture accounted for 11.8% [2].

In both sectors, the main occupational safety hazards are associated with manual handling of loads, repetitive actions, the interaction with machinery and tools, exposure to chemical substances, contact with biological agents, the presence of overhead power lines, and exposure to excessive noise and work at heights [3,4]. In particular,

human–machine interaction represents one of the leading causes of occupational injuries [5] for both agricultural and manufacturing workers. As shown by the latest European statistics, in agriculture, fatal accidents are mainly attributed to tractor-overturn and machinery-related incidents, followed by animal handling [5].

Most deaths involve tractors without or with unfolded rollover protective structures (ROPSs) or misused seatbelts [6]. Whereas, in the manufacturing sector, collisions with falling or moving objects, falls from heights, and being trapped by moving machinery are the leading causes of death [7].

These productive sectors are characterized by the diversity and complexity of the tasks performed. However, while work in the manufacturing sector is highly standardized to increase productivity and create repeatability and logic sequences that reduce errors [8], agricultural activities and tasks vary depending on the type of crop, the number and types of machinery, equipment and tools used, climatic conditions, and daily and seasonal labor requirements [9]. Moreover, poor and unsafe work environments, along with work-related fatigue and stress, pose further occupational safety and health (OSH) threats in both occupational sectors [10].

Most workplace accidents and injuries result from a failure to comply with safety regulations, due to unsafe workers' behaviors and a lack of adoption of proper Personal Protective Equipment (PPE) and protective systems (ROPSs and seatbelts). These behaviors are often indicative of broader systemic deficiencies and hazardous work environments [11]. Stress and organizational factors, such as pressure to meet deadlines, are key influences on unsafe workers' behavior [12].

The current literature emphasizes that workers are not always aware of the role that human behavior plays in contributing to workplace accidents. Some research suggests that workers' familiarity with work tasks/activities can reduce their risk perception due to overconfidence and fatalism. A low-risk perception significantly hinders the adoption of safety behaviors, such as the use of PPE [13] and active participation in safety management practices [12]. Another variable that seems to affect workers' unsafe behaviors is the previous history of accidents. Previous studies conducted in different occupational sectors have reported conflicting results on this factor. Indeed, on the one hand, a group of studies in the industrial and road safety fields [14,15] reported that those who experienced an accident and survived tend to have a lower risk perception and poorer safety behavior, often attributing the cause of such accidents to external factors and not to their unsafe actions. In contrast, other studies [16,17] found a positive relationship between previous involvement in road accidents and an increase in risk perception, resulting in safer traffic behaviors.

Well-structured safety training and effective risk communication are essential to promote safe workplace behaviors while also enhancing workers' commitment to safety and accident prevention. Through safety training, workers can learn to recognize hazards and hazardous actions that must be avoided and understand the consequences of accidents. [18]. Traditional lectures are often considered as passive, particularly for workforces with different sociocultural and professional backgrounds, whereas hands-on training, including field demonstrations and simulations, is considered as more engaging [19]. However, lecture-based training remains the most common method due to its lower cost compared to more engaging approaches [20]. Recent technological advancements have introduced new safety training methods that make the learning process more dynamic and engaging. Studies have highlighted the increasing role of smartphones and mobile applications as flexible training tools, offering cost-effective and easily accessible learning solutions based on the learning-by-doing principle [21]. Mobile applications provide easy access to training materials anytime, anywhere, while offering real-time updates through push notifications and reminders, keeping users informed and engaged [22].

An interactive health and safety platform based on self-directed learning can enhance awareness and effectively influence workers' behavior in real-life situations. Additionally, such tools can support continuous training in small, digestible units, reinforcing knowledge retention and adherence to safety practices [23].

Digital devices with a variety of digital contents, including texts (blog articles, wikis), audios (podcast), videos, graphics (posters), and games, can be used to disseminate information, since they can aid in reaching various target audiences, and promote education, participation, and deeper engagement with the learning materials [24]. Also, these tools' rapid proliferation may be due to the abundance of free and open-source versions available [25]. The use of these types of content, particularly more recent formats such as podcasts and video podcasts, has been previously investigated in the medical and healthcare fields [26]. However, little is known about the interest in using them in other occupational sectors.

Moreover, compared to other listed formats, games can create a more challenging and interactive learning experience. Digital simulations allow users to practice real-world workplace scenarios without actual risk, facilitating safe knowledge acquisition and learning transfer from training to real-life application [27]. Games have already found some applications in the construction industry [28], whereas their application in agriculture remains limited [29]. Similarly, in manufacturing, gamification is still in its early stages, but it is a topic of increasing attention [30].

However, the adoption of digital technologies across different working sectors is influenced by multiple factors. Resistance to change and costs are usually the main barriers that hinder the adoption of digital innovations, while perceived usefulness and ease of use tend to encourage their use. Furthermore, previous studies showed that demographics and farm size in the agricultural context are relevant factors that impact farmers' intention to adopt digital technologies [31]. Similarly, the type of industry or the organization size might be barriers in the industrial sector [32]. On the other hand, increasing competitiveness, the establishment of a safe environment, and improved business image are perceived as driving factors [33].

To increase the adoption of digital innovations, it is fundamental to use devices and create content that can be positively perceived by workers. This approach can not only enhance the usability and acceptance of technologies, but it can also facilitate their more effective integration into daily workflows. The ergonomic User-Centered Design (UCD) approach [34] plays a key role, since it helps emphasize the importance of designing technological solutions based on real users' needs and actively involving users throughout the design process. To our knowledge, the above-mentioned state-of-the-art technologies highlight the need for further investigations, since many digital and technological solutions, especially in the safety training context, still face low adoption rates due to worker resistance and usability challenges. In Italy, the manufacturing and agricultural sectors are among the highest-risk industries in terms of occupational safety. Manufacturing, which in 2021 employed an average of 3.7 million workers [35], recorded 89,674 occupational injuries in the same year, including 208 deaths, mainly due to concussion and internal injuries. In particular, the sectors with the highest number of injuries and fatalities are metal product manufacturing (nearly 22% reported injuries and deaths in the entire manufacturing sector) and machinery manufacturing (13%) [35]. In the agricultural sector, according to the latest Italian national estimates, 872,000 workers were employed, and a total of 26,459 work-related injuries occurred in 2022 alone, with an average of 130 fatal incidents per year, primarily caused by tractor overturns [35]. In Northwest Italy, the Piedmont region exhibits a strong concentration of workers in the metal manufacturing sector, closely linked to the production of transportation equipment and industrial components. Furthermore, the region's agricultural sector is distinguished by a high density of vineyard workers.

Within agricultural operations, viticulture is indeed one of the most prominent and historically significant, contributing largely to cultural and economic value. Vineyards in Piedmont are mainly located on steep slopes, increasing the risk of accidents in the use of mechanized vehicles and enhancing ergonomic risk and musculoskeletal load [36].

Given the high incidence rate of both fatal and non-fatal occupational accidents and work-related diseases in these sectors, Piedmont represents a relevant context in which to investigate issues related to the design of engaging and effective safety training interventions. These two categories of workers are examined together due to their relevance in Northern Italy's production landscape and the need to better support them in adopting safe behaviors and engaging with technological advancements. In addition, identifying differences and shared attitudes between these groups may offer valuable insights for developing safety training guidelines that could be extended to other sectors with similar educational backgrounds or levels of digital literacy.

This study aims to examine the commonalities and differences between two groups of vineyard and manufacturing workers in Piedmont, regarding their perceived working conditions, machinery-related risks, and interest in adopting digital tools for safety training, including their preferred content. Additionally, the study explores the influence of a history of accidents and experience with digital technologies. The results of this investigation will be the foundation for the development of an innovative, digital, and engaging safety training tool by using an ergonomic UCD Approach [34].

2. Materials and Methods

2.1. Study Design

To achieve the research objective, data regarding risk perception, familiarity with digital devices, and preferences regarding safety training were collected through a questionnaire developed based on the existing literature. The study was conducted on a sample from the Piedmont region (Northwest Italy), involving 8 viticulture companies and 1 manufacturing company that offers its clients a comprehensive range of services, including design, construction, assembly, and maintenance of industrial plants across various industries. In particular, the participants from this company operate metalworking machinery and handle its assembly. All of the viticulture companies involved in the study presented similar characteristics in terms of types of terrain (vineyards on steep slopes) and all the participants followed the seasonal workflow of pruning and grape harvesting. The questionnaire was initially tested involving a small sample of workers and then distributed both in a paper and online version. Answers were collected from May 2024 to December 2024. To determine the minimum number of participants required for the statistical tests, a power analysis using G*power software (v. 3.1.9.7) [37] was performed to calculate a priori sample size. Based on this analysis, to uncover a medium effect size (0.65; [38]) with a power of 0.80 and an α level of 0.05, a total sample of 80 participants would be required (40 participants for each group).

The researchers presented the aim of the study during different meetings with managers and workers at each work site. Participants were informed that all data would remain anonymous, and participation was voluntary. After each meeting, those who were willing to participate received the questionnaire and gave their informed consent to be included in the study. No incentives were offered, and participants were free to withdraw at any time. Based on power analysis, responses collection continued until the required number of participants was reached. The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the University of Torino (protocol code 0250477 on 14 May 2024).

2.2. Participants

A total of 79 responses were collected, 40 from vineyard workers and 39 from manufacturing workers. All the participants were males, and their main socio-demographic characteristics are reported in Table 1.

Table 1. Participants' sociodemographic characteristics.

Variable		Viticulture	Manufacturing
		Mean (SD)	
Age (years)		40.05 (18.70)	36.38 (11.58)
Work experience (years)		12.8 (15.14)	15.11 (11.16)
Training courses (N)		2.98 (2.60)	3.23 (1.67)
		N (%)	
Education [1]	Elementary school	6 (15)	0 (0)
	Middle school	16 (40)	14 (35.90)
	High school	16 (40)	23 (58.97)
	University degree	2 (5)	2 (5.13)
Profession	Winegrowers [2]	40 (100)	0 (0)
	Technicians	0 (0)	11 (28.20)
	Assemblers/maintainers	0 (0)	19 (48.72)
	Specialized machine operators	0 (0)	9 (23.08)

[1] The group of manufacturing workers reported a significantly higher level of education, $p < 0.001$, [2] Employed as vineyard workers and not including farmers and entrepreneurs.

2.3. Questionnaire

The questionnaire used in the present study included three main sections. The first section included socio-demographic data, collecting information on gender, age, education, years of work experience, and number of training courses attended in the last five years.

In the second section, participants were asked to rate the following:

(1) Whether they have been involved in different types of accidents in the last 5 years (8 items for vineyard workers and 9 for manufacturing workers, 3-point scale, 0 = never, 1 = once, 2 = more than once) [39]. Then, participants' answers were dichotomized (contrasting the 0 and the other responses) to check whether they had had at least one accident in the last 5 years or not;

(2) Their level of agreement with statements regarding their working conditions and the contribution of human behaviors in leading to accidents (13 items, 4-point scale, 1 = not at all, 2 = a little, 3 = quite a lot, 4 = very much) [40,41];

(3) Whether they operate specific machines and equipment (yes/no question, following [42]) and the perceived level of risk (10 items for vineyard workers and 13 items for manufacturing workers, 4-point scale, 1 = negligible, 2 = low, 3 = medium, 4 = high).

Lastly, the third section examined participants' familiarity and previous experience with digital devices and their perception and preferences regarding digital content for safety training purposes. In detail, the following variables were investigated:

(1) The amount of time participants usually spent per day on various activities using digital devices (from 0 = never, to 5 = more than 2 h) [43];

(2) Interest in adopting digital devices for safety training using a 4-point scale (1 = not interested at all, 2 = not interested, 3 = interested, and 4 = very interested);

(3) Agreement with perceived benefits of using digital devices as a support method for training, on a 4-point scale (1 = not at all, 2 = a little, 3 = quite a lot, 4 = very much) [44];

(4) Perceived importance of incorporating various multimedia content types (e.g., manuals, videos, and games) in a mobile application focused on occupational health and safety, 4-point scale (1 = not at all, 2 = a little, 3 = quite a lot, 4 = very much).

While the overall structure of the questionnaire was the same for both vineyard and manufacturing workers, certain items (e.g., specific machinery used during working days) were tailored to the specific productive industry.

The questionnaire required 10–15 min to be filled out. The questionnaires are reported in Appendix A.

2.4. Data Analysis

Descriptive statistics were computed for all the variables considered. For questions related to conditions and actions that can generally lead to accidents, the means ($1 < \mu < 4$) were used to discern between low and high user perception. Thus, a mean response higher than 2.5 represented a higher level of perception, while a mean response of <2.5 represented a lower level of perception [45].

The internal consistency of the items on the perceived benefits of using digital devices was measured by computing Cronbach's Alpha coefficient. After this reliability analysis, an aggregated score was computed on the considered items as the sum of the responses. Similarly, an aggregated score was computed for the items on the amount of time participants spent per day using digital devices.

Due to the not normally distributed responses, two non-parametric Mann–Whitney tests were performed to compare data between the two groups of workers and between workers who reported accidents and those who did not. Two Spearman correlation analyses, one for each sector, were used to determine the strength and direction of associations. For the aim of the present study, in the correlation analysis, only relevant variables were chosen and commented on. Statistical analyses were performed using IBM SPSS v.29.

3. Results

The following subsections present the descriptive statistics and group differences regarding the following: (i) conditions that lead to accidents and risk perception, (ii) the perceived level of risk of machines and equipment on work sites, and (iii) familiarity and previous experience with digital devices and their perception and preferences. Then, a second results subsection presents the outcomes of the correlation analyses performed on the investigated variables.

3.1. Descriptive Statistics and Differences by Groups

Regarding the experience of previous accidents, a total of 39 workers, of which 18 (46%) were vineyard workers and 21 (54%) workers from the manufacturing sector, reported at least one accident with an injury in the past five years. No statistically significant difference was found between farmers and manufacturing workers.

Regarding the perception of working conditions and the contribution of human behaviors that can lead to accidents, the most critical mean values in both groups were reported for items related to fatigue, need for great concentration and attention, and human error. Additionally, the sample seemed to feel well-informed about safety procedures (Table 2). Mann–Whitney tests showed that vineyard workers perceive significantly higher levels of fatigue, ($U = 367.00$, $z = -4.27$, $p = 0.000$), repute their job as more repetitive ($U = 565.00$, $z = -2.26$, $p = 0.024$) and more defined ($U = 528.00$, $z = -2.64$, $p = 0.008$), have a lot of work to do compared to the available time ($U = 587.50$, $z = -2.05$, $p = 0.041$), have more possibility to decide their work pace ($U = 531.50$, $z = -2.62$, $p = 0.009$), and they perceive a higher level of communication among colleagues ($U = 244.50$, $z = -5.46$, $p = 0.000$). No differences were found comparing items considering workers' previous accident experiences.

Table 2. Analysis of items used to investigate conditions and actions that can generally lead to accidents.

Variable	Viticulture	Manufacturing	Comparison by Group of Workers
	Mean (SD)		*p*
ACC1	2.68 (0.83) [1]	2.41 (0.79)	0.128
ACC2	3.18 (0.75) [1]	2.34 (0.77)	0.000 [2]
ACC3	3.38 (0.67) [1]	3.56 (0,60) [1]	0.189
ACC4	2.13 (1.07)	1.62 (0.82)	0.024 [2]
ACC5	3.25 (0.81) [1]	2.77 (0.81) [1]	0.008 [2]
ACC6	2.70 (0.82) [1]	2.97 (0.71) [1]	0.121
ACC7	2.70 (0.76) [1]	2.33 (0.74)	0.041 [2]
ACC8	2.98 (0.92) [1]	2.44 (0.85)	0.009 [2]
ACC9	2.63 (0.95) [1]	2.64 (0.90) [1]	0.831
ACC10	1.93 (0.66)	1.90 (0.55)	0.981
ACC11	3.05 (0.78) [1]	3.03 (0.63) [1]	0.808
ACC12	3.25 (0.74) [1]	3.28 (0.51) [1]	0.858
ACC13	3.20 (0.65) [1]	2.12 (0.75)	0.000 [2]

[1] mean response higher than 2.5, [2] significant differences.

Regarding the perceived level of risk of machines and equipment on work sites, the sample from the vineyard context reported a higher use of the tractor (24 out of 40) and the pruning shears (21 out of 40). The higher risk was perceived in using the tractor (16 participants out of 24 reported a medium risk and 1 a high risk), the pruning shears (14 out of 21), and the vine shoot shredders (8 out of 14) (Figure 1).

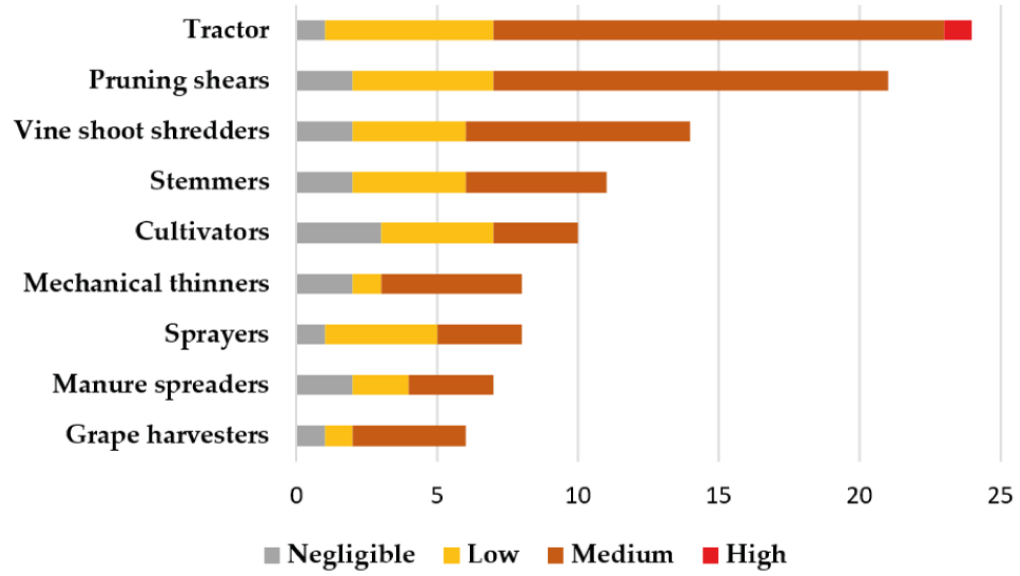


Figure 1. Use and level of perceived risk of the machinery and tools among vineyard workers.

Whereas in manufacturing industries, the forklift, grinding machine and column drill were the most used machineries (21, 20, and 18 out of 39, respectively). However, the risks perceived as higher involved operating the manual lathe (despite the low number of workers that use it, 4 out of 7 perceived a high level of risk), forklift (10 out of 21 reported medium and high level of risk), and guillotine shear (4 out of 9 reported medium and high level of risk) (Figure 2).

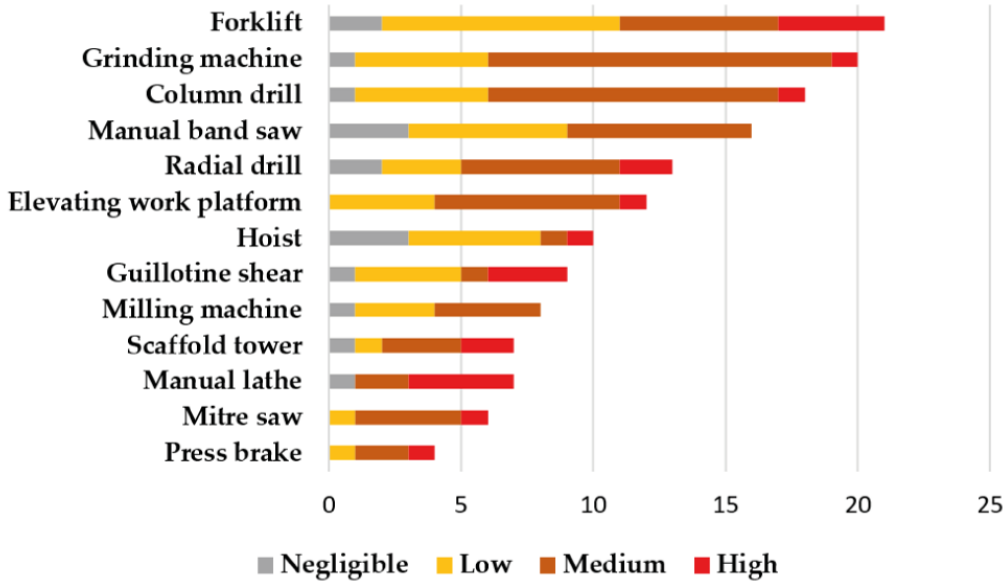


Figure 2. Use and level of perceived risk of the machinery and tools among manufacturing workers.

Regarding the familiarity and experience with digital devices, participants mainly use digital devices for instant messaging and browsing the internet. However, no differences were found between the two groups of workers regarding the time spent on different activities using digital devices and on the aggregated score (Table 3).

Table 3. Analysis of items used to investigate familiarity and preferences regarding digital device contents.

Variable	Viticulture	Manufacturing	Comparison by Group of Workers
	Mean (SD)		*p*
TIME1	2.03 (1.42)	2.15 (1.11)	0.429
TIME2	1.83 (1.39)	1.72 (1.59)	0.476
TIME3	2.98 (1.27)	3.10 (1.43)	0.519
TIME4	3.05 (1.20)	3.42 (1.27)	0.100
TIME5	2.68 (1.51)	2.56 (1.50)	0.730
TIME6	2.35 (1.64)	2.54 (1.54)	0.625
TIME7	0.95 (1.24)	1.10 (1.62)	0.953
INT	2.70 (0.82)	3.05 (0.65)	0.064
BENEFIT1	2.65 (0.83)	3.10 (0.68)	0.014 [1]
BENEFIT2	2.98 (0,80)	3.28 (0.72)	0.087
BENEFIT3	2.65 (0,80)	2.77 (0.74)	0.475
BENEFIT4	2.75 (0,84)	2.87 (0.70)	0.384
CONT1	2.63 (0.90)	3.00 (0.76)	0.053
CONT2	2.70 (0.76)	3.03 (0.81)	0.043 [1]
CONT3	3.10 (0.84)	3.03 (0.78)	0.661
CONT4	2.50 (0.78)	2.67 (0.74)	0.350
CONT5	2.95 (0.88)	2.77 (0.81)	0.344

[1] significant difference.

Despite the mean scores indicating that manufacturing workers seem more interested in adopting safety tools on digital devices if available on the market, no significant difference was reported between the two groups (Table 3).

Regarding the perceived benefits, manufacturing workers perceive safety tools on digital devices as more accessible compared to farmers ($U = 547.00$, $z = -2.45$, $p = 0.014$) (Table 3). No differences were reported for the other items and the aggregated score. When the sums of the scores were computed, the values of coefficient alpha were calculated and were 0.890 and 0.814 for farmers and manufacturing workers, respectively. The values

were considered trustworthy and consistent, given that a coefficient α larger than 0.60 is the requirement for reliability.

When considering the importance of incorporating various types of multimedia content, the higher score for both workers was video lessons. Specifically, among farmers, higher scores were also reported for audio lessons and games, while among manufacturing workers, higher scores were reported for manuals and audio lessons, followed by games (Table 3). A significant difference was found between farmers and manufacturers in audio lessons (U = 588.00, z = −2.02, p = 0.043).

Furthermore, no differences were found when comparing all items in this section considering the workers' previous accident experiences.

3.2. Correlation Analysis

Tables 4 and 5 report correlations for the vineyard and manufacturing workers, respectively. For the aim of the present study, selected variables were chosen, analyzed, and commented on. Hereafter, the main results found for each group of workers and common correlations in both groups will be reported.

Table 4. Correlation of specific selected variables performed for vineyard workers.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1-Age	1													
2-Educ	0.154	1												
3-Train	0.290	0.470 **	1											
4-ACC1	0.228	−0.179	−0.084	1										
5-ACC10	−0.024	−0.412 **	−0.420 **	−0.034	1									
6-ACC11	0.155	−0.110	−0.034	0.443 **	0.207	1								
7-INT	−0.193	−0.141	−0.053	0.030	0.149	−0.025	1							
8-CONT1	0.029	0.052	0.055	−0.106	0.011	−0.222	0.400 *	1						
9-CONT2	−0.443 **	0.042	−0.020	0.013	0.033	−0.046	0.509 **	0.235	1					
10-CONT3	−0.047	−0.075	−0.320 *	0.350 *	−0.036	0.061	0.360 *	0.175	0.557 **	1				
11-CONT4	−0.196	0.039	−0.017	−0.326 *	−0.069	−0.304	0.509 **	0.613 **	0.304	0.093	1			
12-CONT5	−0.384 *	−0.057	−0.064	−0.006	−0.088	−0.023	0.370 *	0.240	0.366 *	0.314 *	0.505 **	1		
13-BENEF [1]	0.077	0.101	0.084	−0.057	−0.083	−0.148	0.512 **	0.359*	0.279	0.364 *	0.369 *	0.305	1	
14-TIME [2]	−0.339 *	−0.142	0.069	−0.125	0.013	−0.132	0.179	0.020	0.168	−0.12	0.400 *	0.516 **	0.196	1

* $p < 0.05$, ** $p < 0.01$, [1] sum of the benefits scores, [2] sum of the scores related to time spent per day on digital devices.

Table 5. Correlation of specific selected variables performed for manufacturing workers.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1-Age	1													
2-Educ	−0.183	1												
3-Train	0.131	0.137	1											
4-ACC1	−0.121	−0.061	0.108	1										
5-ACC10	−0.049	0.148	0.328 *	0.071	1									
6-ACC11	0.154	0.035	0.011	−0.008	0.276	1								
7-INT	−0.164	0.179	−0.136	0.132	0.096	0.003	1							
8-CONT1	0.061	0.191	0.242	0.243	0.174	−0.321 *	0.125	1						
9-CONT2	0.048	0.061	−0.165	0.077	0.123	0.008	0.250	−0.142	1					
10-CONT3	0.045	0.190	0.146	−0.158	0.295	0.145	0.026	0.172	0.386 *	1				
11-CONT4	−0.008	0.003	0.002	0.044	0.036	0.054	0.220	0.247	−0.066	0.111	1			
12-CONT5	−0.273	0.264	0.283	0.050	0.278	−0.039	0.146	−0.192	0.358 *	0.170	0.092	1		
13-BENEF [1]	−0.113	−0.047	−0.017	0.266	0.104	−0.055	0.462 **	0.115	0.047	0.099	0.323 *	−0.073	1	
14-TIME [2]	−0.673 **	0.101	0.01	−0.040	0.129	0.149	0.177	−0.084	−0.182	−0.014	−0.018	0.343 *	−0.062	1

* $p < 0.05$, ** $p < 0.01$, [1] sum of the benefits scores, [2] sum of the scores related to time spent per day on digital devices.

Among the vineyard workers, the belief that most accidents are caused by system failures (ACC10) negatively correlated with the level of education and the number of training courses attended in the last five years. Furthermore, there is a positive correlation between the number of training courses held in the last five years and the perception of system failures as the cause of most accidents. Moreover, the variables concerning the interest in using digital devices to stay updated/informed about health and safety in the workplace (INT) resulted in significant correlations to all the contents suggested (CONT), with preferences for audio/podcast and quizzes/questionnaires. Furthermore,

BENEFIT was positively correlated to the importance they attributed to CONT1, CONT3, and CONT4 as useful content, while in manufacturing, workers only showed a preference for quizzes and questionnaires (CONT4). Vineyard workers' age correlated negatively with the perception of usefulness of audio/podcast (CONT2) and games (CONT5) content for safety training. In both groups, age showed a negative correlation with the time spent on digital devices (TIME), and the perceived benefit (BENEFIT) of adopting digital tools as a safety training method positively correlated with workers' interest (INT). Furthermore, a positive correlation was found between TIME and preference for gaming content in both groups.

4. Discussion

The present study investigated, on the one hand, perceived risks and working conditions, and, on the other hand, familiarity and interest in using digital devices for safety training purposes in a group of vineyard and manufacturing workers in Northwest Italy. The present results will contribute to the development of a new digital safety training tool addressing both sectors.

Overall, it should be acknowledged that in both groups, workers did not exhibit a high-risk perception of their job (ACC1). Given that both sectors are characterized by high levels of occupational risk [1,2], this result may raise some considerations on the need to develop targeted interventions to enhance risk awareness in the workplace. For instance, gamified training proved to be effective [30] in visually explaining the sources of risk and making the nefarious events in the case of incorrect/unsafe behavior visible. However, regarding the working conditions and actions that can lead to accidents, 6 out of 13 considered in the study showed significant differences between the two sectors, indicating that a future digital training tool should have differentiated and targeted content for the two sectors.

Considering perceived fatigue in the workplace, vineyard workers reported significantly higher levels of fatigue compared to manufacturing workers (ACC2). Farmers frequently work for long hours during peak production seasons, leaving them exhausted and sleep-deprived [46]. Such conditions impact their cognitive and physical performances, reducing their attention, memory, reaction time, and executive functions [47]. If in manufacturing breaks are planned and timed, in agriculture rest periods are self-managed and should be encouraged also during training to enhance the awareness of stopping to rest. For instance, in the new digital training tool, some visual elements could appear during farming activity simulations to incentivize trainees to drink, eat fruit, or do some stretching exercises (targeted at hands, neck, arms, etc.) which, if unaccomplished, give a negative score or stop the progress of the game. Having regular rest breaks is indeed known to help maintain performance and reduce fatigue and risks [48].

Regarding the perceived level of work repetitiveness (ACC4), VWs reported a significantly higher score compared to MWs. The results for the VWs are in line with the fact that in agriculture, several highly repetitive job tasks are performed, such as "gripping, high pinch forces, contact stress, and awkward posture associated with the use of non-power hand tools and material handling" [49] as well as repetitive heavy lifting, repetitive movements in awkward postures, such as bending and kneeling, or repetitive movements of the hands and wrists, causing several musculoskeletal disorders [50]. The lower scores reported by the MWs are instead surprising since the manufacturing sector, due to its cycling production characteristics, also requires workers to perform several repetitive tasks, such as the handling of different loads at various frequencies [51]. This difference between the two sectors is probably due to the fact that in manufacturing, the issue of repetitiveness is monitored and benefits have been perceived in the definition of work rhythms, the increase in breaks, and job rotation adoption [48], while in agriculture working and resting

time are self-determined and there is a low work standardization both in every day and seasonal activities [46]. Therefore, the to-be-developed digital training tool could include specific sections or materials to increase farmers' awareness of this specific issue to enhance their self-management of time and work activities.

Clear task allocation is essential for optimizing work efficiency and minimizing the risk of accidents (ACC5). Differences emerged between the two sectors examined in this study regarding the perception of task definition, with manufacturing workers reporting a lower level of agreement. In a manufacturing context, having different types of tasks implies that the skills needed to complete them must also be different [52]. This issue is particularly evident when an insufficient number of personnel are assigned to specific tasks/activities, leading to an increased risk of accidents. Previous studies have indicated that workers have experienced the negative impact of understaffing in shift teams on workplace safety. Thus, workers are compelled to assume greater risks due to the additional workload while production expectations remain unchanged [53].

Vineyard workers reported a higher workload relative to the available time compared to manufacturing workers. The literature indicates that this specific factor contributes to increased levels of stress, fatigue, and burnout [47]. Despite this, the vineyard sample, on average, perceived a greater ability to regulate their own working time compared to manufacturing workers (ACC7). This difference within the manufacturing sector can be attributed to the fact that, in manufacturing, the work rhythms and work pace are strictly connected with productivity, requiring complete the planned production under time pressure [54]. Therefore, to accelerate task accomplishment, workers may overload transport equipment, conduct maintenance on operational machinery without halting production to maintain workflow continuity, neglect communication with colleagues, or lift heavy loads improperly, thereby increasing the risk of injury [53]. Moreover, workers may struggle with their tasks if the preceding shift team leaves any unfinished work, further exacerbating workplace pressure [53].

The level of communication within work groups was found to be higher among VWs compared to MWs, a finding consistent with previous results in the viticultural context, where satisfaction with internal communication was relatively high [55]. In manufacturing, previous studies have mainly focused on communication between employers and employees [56], while few have considered communication between workers [57]. However, the low score reported in the present study among MWs can be explained by [58], whose study acknowledges that machine operators are rarely engaged with their colleagues as they focus primarily on ensuring the proper functioning of the machines, with limited interaction with other operators. Oral communication with peers among farmers is mostly used to share knowledge and information about innovation and diversified agricultural practices [59], while in manufacturing the weak motivation may be due to a more repetitive interaction with the machine that requires behaviors transmitted through specific training. Communication between colleagues occurs only when necessary [58]. Peer interaction, when incorporated into a training session, can be useful for enhancing socialization among co-workers during breaks and increasing social well-being in the workplace. In certain contexts, it may also encourage discussions and knowledge-sharing to promote appropriate and safe behaviors, which tend to happen more naturally in face-to-face settings [60]. Further exploration is needed to determine how to implement this effectively in digital training solutions.

Regarding the perceived risks in the use of specific machinery in the workplace, findings indicate that nearly the entire vineyard sample did not consider any equipment to have a high degree of risk. This phenomenon can be explained by the concept of optimism bias, which refers to the tendency to perceive negative events as less likely to affect oneself

compared to one's peers [61]. Among vineyard workers, the two tools perceived as the most hazardous were tractors and pruning shears. This perception aligns with accident data in the agricultural sector, where tractors have been identified as the leading cause of fatal accidents [6], while pruning shears rank among the tools responsible for the highest number of non-fatal injuries [62].

The operation of industrial machinery poses significant hazards for workers responsible for overseeing manufacturing productive processes. Despite the fact that several safety measures are usually implemented, including protective gear, laser beams, and safety cabs, some direct hazards persist [63]. Such hazards include clothing entanglement in moving parts, the absence or improper use of safety guards on drills, and exposure to flying chips and waste from lathes [64]. Injuries associated with lathe operation are particularly relevant, accounting for a larger proportion of machine-related accidents in this sector, primarily due to the high-speed rotation of its components [65] and accounting one-tenth of the accidents reported in this industry [66]. Furthermore, machinery with sharp edges, such as presses, are still one of the main causes of amputations, which constitute 60% of such injuries [67]. However, similarly to the construction sector, working at heights remains a leading cause of severe and fatal accidents also in manufacturing. Falls from height are among the primary contributors to occupational injuries and fatalities [5], highlighting the critical need for enhanced safety measures in high-risk work environments.

In this study, no significant differences were observed in perceived working conditions and risks between participants who had experienced an accident and those who had not. This result contributes to the current debate on this topic [14–17] and is consistent with the results of a study conducted among foundry workers, which found no relationship between previous involvement in occupational accidents and risk perception [68]. This evidence also highlights that in the development of the new safety training tools, there is no need to take workers' previous history of accidents into account to improve workers' safety behaviors.

The reported level of interest in using digital devices for safety training tended to be quite high for both groups, and even though MWs reported a slightly higher mean, no significant differences were found. This evidence is encouraging for the expected acceptability of the new digital training tool, and it is in line with a recent review [69,70], showing how the confidence in technologies (e.g., extended reality) to improve safety training outcomes can enhance safety training effectiveness across various high-risk industries.

No differences were found between the two groups of workers regarding the time spent on different activities using digital devices, and, consistent with the literature [71,72], the use of digital devices during the day negatively correlated with age in both samples. This result points out a relevant challenge when designing a digital safety training tool for workers when considering the aging of the workforce [73].

Moreover, the time spent daily on digital devices positively correlated with the preference for gaming content for safety training purposes. To our knowledge, there are no studies in the literature that have highlighted such an association; however, this could be explained by the influence of the age variable, which correlated in the vineyard sample with the preference for game contents and with time spent on digital devices. Nevertheless, for both groups, the preferred multimedia content to convey safety information was video lessons. Furthermore, numerous studies investigated the effectiveness of video lessons even in occupational safety contexts [74]. The resulting suggestion is that in the development of the new safety training tool, video lessons should always be included, even if combined with another form of training (lectures, e-learning, or gamified).

Regarding the level of education, it did not correlate with the other variables investigated for MWs. However, in the sample of VWs, education level was negatively correlated

with the perception that most workplace accidents are caused by system failures. This is consistent with Kouabenan [16], who suggested that individuals with lower levels of education may be more fatalistic and believe that external factors, rather than their own behaviors, are the cause of negative events and accidents.

Some limitations should be acknowledged in the present study. The first one is related to the gender issue, since all workers who took part in the study were males and we could not investigate any gender differences. Our sample was consistent with the current Italian context, where the sectors under investigation remain predominantly male-dominated [35,75]. However, since recent statistics at both the European and global levels report a significant increase in women's employment within manufacturing and agriculture occupational sectors [76,77], future research should carefully consider the involvement of gender-balanced samples to detect similarities and differences.

The second limitation of this study is related to the small and region-specific sample. Despite being sufficient according to the power analysis and in line with previous investigations in the agricultural and manufacturing sectors [78,79], we acknowledge that the small number of participants limits the generalizability of the results. Even though the short-term implications of our results may be at the local level, the methods and the process of analysis may be replicated involving larger samples in other geographical areas to provide insights for the development of innovative and engaging safety training that considers similarities and differences between the occupational contexts.

No significant differences were identified between the two groups in terms of sociodemographic characteristics, thereby enhancing their comparability from a statistical perspective and allowing us to deduce common attitudes. Even though the VWs had a significantly lower level of education, it should be highlighted that almost no items correlated to the level of education and there were no differences between the two groups. This result points out that risk perception and attitudes toward technology may be addressed with safety training in our sample regardless of education. Only in the VW sample did the belief that most workplace accidents are caused by system failures (ACC10) correlate with education. The results involving this variable should be taken cautiously. Future investigations and the involvement of a wider sample may solve this issue.

Finally, about future research developments, considering the benefits that digital devices and their content types (videos, texts, podcasts/audio, and games) can offer, in addition to their accessibility, investigations may be conducted on their design as a valuable resource for foreign workers. This becomes even more relevant when considering the increasing rate of migrant workers employed in Italy in different sectors [80]. Digital training tools may help overcome language barriers through the extensive use of visual elements or by providing real-time translations, leveraging the advancements in AI. Future research should investigate these aspects when analyzing the design of digitalized training.

In addition, given the exploratory quantitative nature of this research, aimed at identifying sectoral trends and training preferences, future studies would benefit from a qualitative approach to substantiate the quantitative results and understand why these trends occur, and to enhance the understanding of workers' motivations, experiences, and barriers to adopting innovative digital safety solutions.

5. Conclusions

Despite working in the two most hazardous sectors in Italy, both the vineyard and the manufacturing participants reported a medium level of overall perception of working conditions and actions that can lead to accidents. Only a specific and limited amount of machinery and equipment (such as tractors and electric shears for viticulture and lathe and

forklift for the manufacturing industry) were perceived as relevant sources of risk during interaction.

Regarding the use of digital devices as tools for safety training, the participants showed quite high levels of interest, positively suggesting the willingness of workers to approach new and alternative training methods. On the other hand, the total time spent during the day using digital devices positively correlated with the participants' age. The most preferred training tool among both samples was video lessons, but gamification was the most appreciated content by individuals who spend the most time using digital devices. This suggests that safety training and the promotion of safe behaviors may benefit from an increasing orientation towards gamification, particularly for today's young adult generations. Such an approach could serve as an engaging and complementary method alongside theoretical instruction, even when delivered through recorded video lessons.

This study contributes to the field offering initial suggestions on the design of safety training content clearly showing that sector-specific working conditions and workers' risk perception can influence the development (and the subsequent effectiveness) of digital safety training tools. To ensure their success, training content must be carefully calibrated to reflect the demands, routines, and behavioral patterns of each sector.

For instance, for vineyard workers, digital training should be designed to represent their real daily work, which is characterized by high physical fatigue, repetitive tasks, the need for sustained concentration limited task standardization, and a heavy workload relative to the time available and the need to self-regulate their work pace for their well-being. In contrast, manufacturing workers, who work in a more structured work environment, may benefit more from scenario-based training modules that emphasize task clarity and reinforce peer communication. Aligning training content with these sector-specific factors can lead to more effective, engaging learning experiences.

Furthermore, although both groups of workers showed interest in digital safety tools, only young VWs expressed a preference for gamification, while for all the other participants, preferences about digital safety training content remained limited to their routinary device use. A user-centered design of digital safety training can also support the seamless integration of gamified content into daily routines, making the tools appealing and usable for older and younger workers in different occupational contexts.

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Abbreviations

The following abbreviations are used in this manuscript:

VW	Vineyard workers
MW	Manufacturing workers
UCD	User-centered design
SPSS	Statistical Package for Social Science
INT	interested in adopting safety tools
CONT	content

Appendix A

Table A1. Questionnaire.

Question	Item	ID Code	Scale
	SECTION 1:		
sociodemographic information	Select your gender		1 = male, 0 = female, 2 = other
	How old are you?		Open question
	What is your educational qualification		0 = none, 1 = primary school, 2 = middle school, 3 = high school, 4 = degree, n.a. = I prefer not to answer
	How long have you been working in this field?		Open question
	How many training courses on workplace safety have you attended in the last 5 years?		Open question
	SECTION 2:		
Have you had accidents during work activity resulting in damage in the last 5 years?	For vineyard workers: Fall from machinery/tractor, fall from height, slipping, accident with tractor, accident with other machinery, accident with other equipment, struck, environmental For manufacturing workers: crushed by a vehicle, rollover, run over a vehicle, struck by falling objects, fall from height, electrocution, inhalation of fumes, ocular foreign body, other road accidents		0 = never, 1 = once, 2 = more than once
Thinking about the work you do, please indicate how true you consider the following statements about working conditions, on a scale from 1 to 4	My job is dangerous	ACC1	1 = not at all, 2 = a little, 3 = quite a lot, 4 = very much
	My job is tiring	ACC2	
	My job requires great concentration and attention from me	ACC3	
	My job is repetitive and does not involve alternating with other tasks or activities	ACC4	
	My tasks are well-defined	ACC5	
	There is enough manpower to complete the daily work	ACC6	
	There is too much work to do compared to the available time	ACC7	
	I have the possibility to decide my work pace	ACC8	
	My earnings are proportional to my work	ACC9	
	Most accidents are caused by system failures	ACC10	
	Most accidents are caused by human behavior	ACC11	
	I feel well-informed about safety procedures	ACC12	
	There is good communication within my work group	ACC13	
Which of the following tools and machinery do you use in your work activity? Please indicate, according to you, the level of risk for each tool and machinery that you use, on a scale from 1 to 4	For vineyard workers: tractor, pruning shears, cultivators, vine shoot shredders, mechanical thinners, manure spreaders, sprayers, grape harvesters, stemmers. For manufacturing workers: manual lathe, grinding machine, column drill, radial drill, milling machine, guillotine shear, press brake, manual band saw, miter saw, forklift, scaffold tower, hoist, elevating work platform.		1 = negligible, 2 = low, 3 = medium, 4 = high

Table A1. *Cont.*

Question	Item	ID Code	Scale
SECTION 3:			
Please indicate how much time on average you dedicate per day to each of the following activities on digital devices	Reading (e.g., newspapers, blogs, etc.) Sending and receiving emails Using instant messaging apps (e.g., WhatsApp) Browsing the internet Watching/listening to videos/music Checking social media (e.g., Instagram, Facebook) Playing games	TIME1 TIME2 TIME3 TIME4 TIME5 TIME6 TIME7	0 = never, 1 = a few minutes, 2 ≤ 30 min, 3 = 30 min to 1 h, 4 = 1–2 h, 5 = more than 2 h
	Please indicate how interested you would be/are in using digital devices to stay updated/informed about health and safety in the workplace	INT	1 = Not at all 2 = A little 3 = Enough 4 = A lot
Please indicate how much you agree with the following statements on a scale from 1 to 4: The use of platforms on digital devices as a method of support for training can make safety training...	more accessible faster more interesting more effective	BENEFIT1 BENEFIT2 BENEFIT3 BENEFIT4	1 = not at all, 2 = a little, 3 = quite a lot, 4 = very much
If you were to use digital devices as a method of support for training, how important do you think each of the following contents should appear on a scale from 1 to 4	Manual/documentation (e.g., written documentation on procedures and risks) Audio lessons/podcasts Video lessons Practice exercises in the form of quizzes Practice exercises in the form of games (e.g., quizzes with scores and rankings)	CONT1 CONT2 CONT3 CONT4 CONT5	1 = not at all, 2 = a little, 3 = quite a lot, 4 = very much

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